Notes: Iaria, Schwarz, and Waldinger (QJE, 2018)

Frontier Knowledge and Scientific Production: Evidence from the Collapse of International Science

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1 Big picture

- **Question.** How much does access to frontier knowledge matter for scientific progress and for the later development of new technology?
- **Setting.** The paper uses World War I and the postwar Allied boycott of Central scientists as a large shock to international scientific cooperation.
- **Core empirical strategy.** First, document that the war and boycott sharply reduced international knowledge flows. Second, compare productivity changes of scientists in field-country pairs that relied on frontier research from abroad with productivity changes of scientists in field-country pairs that relied on frontier research from home.
- **First-stage historical fact.** The scientific world split into Allied and Central camps. International journals were delayed, conferences collapsed, and Central scientists were excluded from many postwar scientific organizations and meetings.
- **First-stage empirical fact.** After 1914, papers cite much less recent foreign research from outside the camp, and paper titles become less similar across enemy camps. Both patterns indicate a sharp reduction in international knowledge flows.
- **Main second-stage finding.** Scientists who relied on frontier research from outside the camp published fewer papers, produced less Nobel Prize-nominated research, introduced fewer novel scientific words, and introduced fewer novel words that later appeared in patents.
- **Magnitude.** The share of citations to recent research from outside the camp falls by about 0.22 relative to home citations, an 85% decline relative to the prewar share. The relative similarity of paper titles to papers from outside the camp falls by about 0.47 standard deviations. In the productivity analysis, access to the top 1% frontier matters much more than access to the top 3% or top 5%.
- **Illustrative economic interpretation.** U.S. biochemists, who relied heavily on top German frontier research, experience much larger post-1914 productivity declines than U.S. biologists, who relied more on home frontier research.
- **Main conceptual contribution.** The paper gives direct evidence that access to the very best existing knowledge is a key input into the production of new science and into the later creation of new technology.
- **Why we care.** The paper links openness in scientific exchange to both basic scientific output and technological application, and it does so with unusually rich historical data.

2 Data and measurement (key objects)

2.1 Historical censuses of university scientists

- The paper digitizes two worldwide censuses of university scientists from *Minerva - Handbuch der Gelehrten Welt*.
- It digitizes more than 2,500 pages and more than 60,000 individual records overall.
- Coverage:
  - 569 universities in 1900,

  - 973 universities in 1914,
  - 23,917 professors in 1900,
  - 36,777 professors in 1914.
- The analysis focuses on five scientific fields where journal publication was already standard: medicine, biology, chemistry, physics, and mathematics.
- In those five fields the data contain:
  - 10,133 scientists in 1900,
  - 15,891 scientists in 1914.

2.2 Publication and citation data

- The paper collects all papers published in 160 top scientific journals from ISI Web of Science for 1900-1930.
- It also collects cited references and citation counts for those references.
- The key challenge is that most historical journals do not report author affiliations. The paper assigns countries to authors and cited references using a three-step procedure:
  - affiliations when journals report them,
  - country information from the 1900 and 1914 scientist censuses,
  - propagation of country information within journal-author-name combinations.
- Country assignment is fractional for coauthored work. A paper jointly written by one U.S. and one U.K. author counts as 0.5 U.S. paper and 0.5 U.K. paper.

2.3 Nobel and patent data

- Nobel data come from the Nobel archives and cover nominations in physics, chemistry, and physiology/medicine from 1905-1945.
- There are 993 nominees in total, 131 of whom eventually win.
- Patent data consist of the full text of more than 2.5 million U.S. patents from 1920-1979, containing more than 7.5 billion words.
- The patent text is used to trace whether newly introduced scientific words later appear in patents.

2.4 Final analysis datasets

- **Paper-level dataset.** Used for citation-share and title-similarity analysis. It includes papers for which the country of at least one author and one reference can be matched.
- **Scientist-level panel.** Used for productivity analysis. It includes all university scientists who published at least one paper between 1905 and 1930.

2.5 Key measurement choices

- **Recent research.** References to research published in the preceding five years.
- **Frontier research.** Research that ends up in the top 5%, top 3%, or top 1% of the subject-level citation distribution, counting citations until today.
- **Citation shares.** For each paper, references are split into research from home, foreign countries inside the camp, and foreign countries outside the camp.
- **Title similarity.** The paper applies LSA to translated paper titles and then computes cosine similarity between titles.
- **Novel words.** Words first introduced by a scientist in a paper title between 1905 and 1930 that had not appeared in any earlier paper title in the 160-journal corpus.
- **Patent-relevant words.** Novel scientific words weighted by the number of times they appear in later U.S. patent texts.

2.6 Useful summary statistics

- The average paper contains 17.6 references overall, 7.4 references to recent research, and 4.6 references to recent research within the 160-journal sample. Of those 4.6, the paper can match the country for about 3.0 references, and citation counts for about 2.6 references.
- In the matched citation sample, about 68.6% of recent references cite research from home, 15.9% cite foreign research inside the camp, and 15.0% cite foreign research outside the camp.
- For top 1% frontier research, the average recent-reference shares are 5.4% from home, 1.2% from inside the camp, and 1.3% from outside the camp.
- The scientist-level panel contains 8,734 scientists and 227,084 scientist-year observations.
- In that panel, the average scientist produces 0.267 publications per year, 0.001 Nobel-nominated papers per year, 0.041 novel words per year, and 0.427 patent-relevant novel words per year.

3 Models and empirical specifications (all equations + interpretation)

3.1 Citation-share setup

For each citing paper, let c^{Home}, $c^{\text{Foreign-IN}}$, and $c^{\text{Foreign-OUT}}$ denote counts of recent citations to research from home, from foreign countries inside the camp, and from foreign countries outside the camp. Dividing by total references C^{Total} yields the three citation shares.

Figure III illustrates the construction with a simple example: a U.S. paper with four recent references, two to U.S. research, one to German research, and one to U.K. research has shares

$$\frac{c^{\text{Home}}}{C^{\text{Total}}} = \frac{2}{4} = 0.5, \qquad \frac{c^{\text{Foreign-IN}}}{C^{\text{Total}}} = \frac{1}{4} = 0.25, \qquad \frac{c^{\text{Foreign-OUT}}}{C^{\text{Total}}} = \frac{1}{4} = 0.25.$$

3.2 Baseline citation-share regression

The baseline regression is

$$
\begin{aligned}
Y_{ic} = {} & \omega_1 \mathbf{1}\{c = \text{Foreign Out}\} + \omega_2 \mathbf{1}\{c = \text{Foreign Out}\}\mathbf{1}\{t(i) = \text{Post-1914}\} \\
& + \iota_1 \mathbf{1}\{c = \text{Foreign In}\} + \iota_2 \mathbf{1}\{c = \text{Foreign In}\}\mathbf{1}\{t(i) = \text{Post-1914}\} \\
& + \text{Citing Paper FE}_i + \varepsilon_{ic}.
\end{aligned} \tag{1}
$$

- **Unit of observation.** Three observations per citing paper: home, foreign-inside-camp, and foreign-outside-camp.
- **Dependent variable.** Citation share to one of the three source categories.
- **Omitted category.** Citation share to home research.
- ω_2**.** Change after 1914 in citations to research from outside the camp, relative to home.
- ι_2**.** Change after 1914 in citations to research from foreign countries inside the camp, relative to home.
- **Why paper fixed effects matter.** They absorb any paper-level tendency to cite more or fewer papers and implicitly absorb country, field, and time differences that are common within the paper.

3.3 Year-by-year citation dynamics

To study timing, the paper estimates

$$
\begin{aligned}
Y_{ic} = {} & \sum_{\tau=1905}^{1930} \omega_\tau \mathbf{1}\{c = \text{Foreign Out}\}\mathbf{1}\{t(i) = \tau\} \\
& + \sum_{\tau=1905}^{1930} \iota_\tau \mathbf{1}\{c = \text{Foreign In}\}\mathbf{1}\{t(i) = \tau\} \\
& + \text{Citing Paper FE}_i + \varepsilon_{ic}.
\end{aligned} \tag{2}
$$

- **Interpretation.** This produces an event-study style path for the relative foreign citation shares.
- **Use.** It shows when citation shares collapse and whether they later recover.

3.4 Frontier-research citations

The same regression logic is applied after splitting references into frontier and nonfrontier research, where frontier is alternatively defined as top 5%, top 3%, or top 1% of the eventual citation distribution.

- **Interpretation.** This asks whether the war mainly cut scientists off from ordinary foreign research, or whether it also cut them off from the very best research abroad.
- **Key idea.** If the top 1% results are especially large, access to the thin edge of the frontier matters disproportionately.

3.5 Latent semantic analysis and title similarity

The paper applies LSA to translated paper titles. After stemming and removing stopwords, the title corpus contains 79,438 paper titles and 35,119 unique stems.

The cosine similarity between documents i and j is

$$\cos(i,j) = \frac{\sum_{c=1}^{C} \delta_{i,c}\delta_{j,c}}{\sqrt{\sum_{c=1}^{C} \delta_{i,c}^2}\sqrt{\sum_{c=1}^{C} \delta_{j,c}^2}},$$

where $\delta_{i,c}$ is the cth component of document i in the reduced LSA representation.

For each paper in year t, the paper computes:

- similarity to the most similar paper from home,
- similarity to the most similar paper from inside the camp,
- similarity to the most similar paper from outside the camp,

using papers published between $t-4$ and t. It also computes the average similarity to the five most similar papers in each camp.

- **Interpretation.** This is a text-based measure of whether research programs remain aligned across camps.
- **Why useful.** Citation behavior could reflect strategic citation choices or politics. Title similarity provides an independent measure based only on the content of titles.

3.6 Scientist productivity regression

The main productivity regression is

$$\begin{aligned} Y_{ift} = {} & \beta_1 \big(\text{Prewar Reliance on Frontier OUT}\big)_{if} \mathbf{1}\{t = \text{Post-1914}\} \\ & + \beta_2 \big(\text{Prewar Reliance on Frontier IN}\big)_{if} \mathbf{1}\{t = \text{Post-1914}\} \\ & + \text{Scientist FE}_{if} + \text{Year FE}_t + X_{ift}\theta + \varepsilon_{ift}. \end{aligned} \tag{3}$$

- **Unit of observation.** Scientist-year.
- **Dependent variable.** First, yearly publication counts in top journals; later, alternative outcomes such as Nobel-nominated papers, novel words, and patent-relevant words.
- **Treatment intensity.** Prewar dependence on frontier knowledge from outside the camp or from foreign countries inside the camp, measured at the field-country level using pre-1914 citation shares.
- **Omitted category.** Prewar reliance on frontier research from home.
- β_1**.** Post-1914 differential change in output for scientists in field-country pairs that relied more on frontier knowledge from outside the camp.
- β_2**.** Same idea for frontier knowledge from foreign countries inside the camp.
- **Controls.** The regressions include scientist fixed effects, year fixed effects, nonfrontier-reliance controls interacted with post-1914, and career-age-by-field interactions.

3.7 Year-by-year productivity dynamics

The event-study version is

$$
\begin{aligned}
Y_{ift} = & \sum_{\substack{\tau=1905 \\ \tau\neq 1913}}^{1930} \beta_{1\tau} \big(\text{Prewar Reliance on Frontier OUT}\big)_{if} \mathbf{1}\{t=\tau\} \\
& + \sum_{\substack{\tau=1905 \\ \tau\neq 1913}}^{1930} \beta_{2\tau} \big(\text{Prewar Reliance on Frontier IN}\big)_{if} \mathbf{1}\{t=\tau\} \\
& + \text{Scientist FE}_{if} + \text{Year FE}_t + X_{ift}\theta + \varepsilon_{ift}.
\end{aligned} \tag{4}
$$

- **Purpose.** Check pretrends and show the timing of the productivity collapse.
- **Interpretation.** If pretrends are flat or favorable before 1914 and the lines break sharply afterward, that supports the causal interpretation.

3.8 Alternative outcome construction

- **Nobel-nominated paper.** For each Nobel nominee, the paper identifies the highest-cited paper published before the nominee's last nomination and treats that paper-year as Nobel-nominated research.
- **Nomination-weighted outcome.** The Nobel-nominated indicator is weighted by the number of nominations received in the final two years before the last nomination.
- **Novel words.** Count of words first introduced by a scientist in a title between 1905 and 1930, excluding frequently used general words and numbers.
- **Patent-relevant words.** Each novel word is weighted by the number of times it appears in U.S. patents granted between $t + 15$ and $t + 30$.

4 Identification and instruments (what makes the estimates credible)

4.1 Core identifying idea

- The first-stage exposure shock is historical and geopolitical: WWI plus the Allied boycott suddenly interrupted scientific exchange across the Allied-Central divide.
- The second-stage exposure design compares *differential* effects across field-country pairs depending on how much they relied on frontier research from abroad before the shock.
- This is a shift-share-style difference-in-differences design:
  - the common shock is the collapse in international scientific cooperation after 1914,
  - the exposure shares are prewar reliance on frontier research from home, from inside the camp, and from outside the camp.

4.2 Why the first-stage shock is real

- Journal deliveries were delayed dramatically, especially for enemy-country journals.
- International conferences nearly disappeared during the war.

- Central scientists were excluded from many postwar scientific meetings and organizations until 1926.
- Citations to recent foreign research collapsed, and title similarity across enemy camps declined sharply.

4.3 Why the citation evidence is interpreted as knowledge-flow disruption rather than political citation bias

- The paper shows that citations to *prewar* foreign research do *not* fall disproportionately after 1914.
- If scientists still knew the old foreign work and continued citing it, but stopped citing *recent* foreign work, the natural interpretation is that they were cut off from recent knowledge rather than merely refusing to cite for political reasons.
- Additional evidence: neutral countries do not show the same collapse in citation patterns, and excluding chemists does not make the results disappear.

4.4 What the fixed effects absorb

- **Citing paper fixed effects** absorb paper-level citation propensities in the citation-share regressions.
- **Scientist fixed effects** absorb permanent scientist quality in the productivity regressions.
- **Year fixed effects** absorb common time shocks such as the general wartime productivity decline.
- **Camp-year, field-year, and camp-field-year fixed effects** absorb increasingly fine common shocks, including the possibility that Allied chemists or Central physicists are hit by special time-varying conditions.

4.5 Robustness logic

- The paper adds country-level war intensity controls: combat on own territory, total deaths per capita, and civilian deaths per capita.
- It collects more than 6,500 obituaries to show that scientists in the sample, and especially scientists reliant on frontier research from abroad, do not die disproportionately during WWI.
- It excludes chemists, the group most involved in wartime weapons work.
- It restricts publication outcomes to own-camp journals, to show the results are not just a publication-opportunity artifact.
- It re-estimates within the United States only, where the war is not fought on domestic territory and where a rise of U.S. science cannot mechanically explain the differential field-level pattern.

4.6 Interpretation of the productivity design

- Scientists are not directly "treated" at the individual level. Instead, scientists in different field-country cells have different *prewar exposure* to frontier knowledge from abroad.

- The identifying comparison is therefore: after the common post-1914 collapse in international science, do high-foreign-frontier fields decline more than high-home-frontier fields?
- Because exposure is measured before the war, reverse causality from postwar productivity to exposure is not a concern.

5 Tables and figures

5.1 Table I: Scientific camps during WWI and the boycott

Read:

- This is the institutional map of the paper. It defines Allied, Central, and Neutral countries exactly as used in the empirical analysis.
- The paper's comparison is fundamentally about the breakdown of knowledge flows *across* the Allied-Central divide.
- The table also reminds you that the "inside camp" versus "outside camp" distinction is historically meaningful rather than arbitrary.

TABLE I

SCIENTIFIC CAMPS DURING WWI AND THE BOYCOTT

Allies	Centrals	Neutrals
United States	Germany	Switzerland
United Kingdom (incl. Ireland)	Austria	Netherlands
France	Hungary	Sweden
Canada	Bulgaria	Denmark
Japan	Ottoman E./Turkey	Norway
Italy		Czechoslovakia
Belgium		Finland
Australia		Spain
Romania		Monaco
Poland		
Brazil		
South Africa		
Greece		
New Zealand		
Portugal		
Serbia		

Notes. Countries are classified following the definition of the International Research Council (IRC) and ordered by scientific output in our data. Austria-Hungary was split into two countries after WWI. Czechoslovakia was part of Austria-Hungary before WWI and became a sovereign state after 1918. Turkey emerged from parts of the Ottoman Empire after WWI.

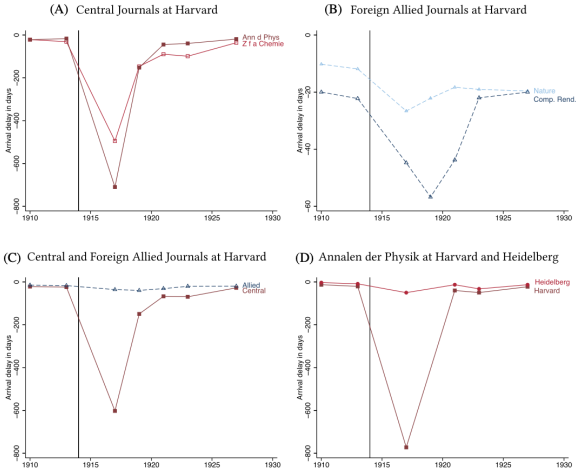

FIGURE I

Arrival Delay of International Journals

Panel A plots the average delay between publication and arrival date at the Harvard library for the German journals *Zeitschrift für Analytische Chemie* and *Annalen der Physik*. Arrival dates are based on library entry stamps (see Online Appendix Figure A.1 for an example). Delays are calculated as yearly averages for 1910, 1913, 1917, 1919, 1921, 1923, and 1927. Panel B plots the delay for two Allied journals, the British journal *Nature* and the French journal *Comptes Rendus*. Panel C compares average delays for German journals and Allied journals. Panel D compares delays for the *Annalen der Physik* at Harvard and at the University of Heidelberg in Germany. In Panel D, the delay at Harvard differs slightly from the delay reported in Panel A because we focus on journal issues that were available both at Harvard and at Heidelberg. Data on entry stamps were collected by the authors at Harvard and at the University of Heidelberg (see Online Appendix E.1 for details).

5.2 Figure I: Arrival delay of international journals

Read:

- German journals at Harvard suffer an enormous wartime delay. The *Zeitschrift fur Analytische Chemie* and *Annalen der Physik* go from roughly 26 days of delay before the war to about 500 days in 1917, then recover only slowly.

- Allied journals are also delayed, but much less: the French *Comptes Rendus* goes from about 21 days to about 57 days in 1919, and *Nature* goes from about 10 days to about 27 days during the war.
- Comparing Harvard with Heidelberg shows that the delay for German journals at Harvard is much larger than the delay within Germany, so this is not just a German publishing breakdown.
- The figure gives concrete evidence that foreign scientific knowledge - especially from the enemy camp - became physically harder to access.

5.3 Table II: Attendance of International Congresses of Mathematicians

Read:

- Germany sends large delegations before the war, but zero delegates in Strasbourg (1920) and Toronto (1924).
- The 1916 Stockholm congress is canceled entirely.
- By 1928 in Bologna, Germany returns with 106 delegates, indicating that the boycott has effectively ended.
- This table is a clean quantitative summary of the boycott's effect on elite international scientific exchange.

TABLE II

ATTENDANCE OF INTERNATIONAL CONGRESSES OF MATHEMATICIANS

Year	Location	Delegates from:							
		Germany	Switzerland	France	U.S.	Canada	U.K.	Italy	Others
1897	Zürich	53	68	29	7	0	3	25	57
1900	Paris	26	7	93	19	1	12	23	69
1904	Heidelberg	204	13	29	19	1	8	14	108
1908	Rome	174	18	92	27	1	33	213	142
1912	Cambridge (U.K.)	70	10	45	87	5	270	41	181
1916	Stockholm	Canceled							
1920	Strasbourg	0	12	112	15	1	11	7	99
1924	Toronto	0	5	45	270	118	93	15	80
1928	Bologna	106	48	91	76	7	64	412	312
1932	Zürich	142	185	89	102	2	49	81	203

Notes. The Table reports the number of delegates at each *International Congress of Mathematicians*. Data were collected by the authors from historical issues of *Proceedings of the International Congresses of Mathematicians* (see Online Appendix E.2 for details).

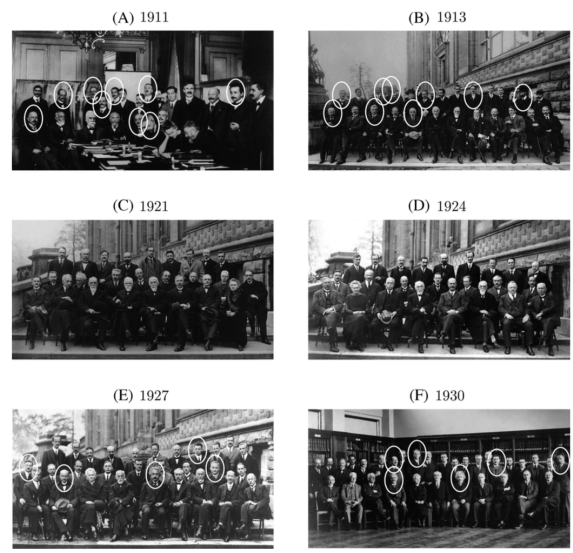

FIGURE II

Central Attendance at Solvay Conference

The figure shows delegates at the Solvay Conferences in physics. Circles indicate delegates from Central countries. See Online Appendix Table A.2 for delegate names. Data were collected by the authors from Mehra (1975) (see Online Appendix E.3 for details).

5.4 Figure II: Central attendance at Solvay Conference

Read:

- The Solvay Conferences were an ultra-elite venue in physics. Before the war, Central scientists are well represented: 9 of 24 participants in 1911 and 9 of 31 in 1913.

- During the war, the conferences stop. In 1921 and 1924 Central scientists are excluded; by 1927 they return.
- Because Solvay is one of the most important scientific conferences in history, this figure shows that the collapse in international cooperation reaches the very top of the scientific frontier.

5.5 Table III: Summary statistics for citation shares and title similarity

Read:

- The baseline citation structure is already home-biased: about 68.6% of recent citations are to home research, versus 15.9% to foreign-inside-camp research and 15.0% to foreign-outside-camp research.
- For top 1% frontier research, home still dominates: 5.4% of all references cite top 1% home research, versus only 1.2% inside camp and 1.3% outside camp.
- The LSA measures show the same qualitative ranking: papers are more similar to home papers than to foreign papers.

TABLE III

SUMMARY STATISTICS: INTERNATIONAL CITATION SHARES AND LSA TITLE SIMILARITY

	(1)	Home (2)	Foreign inside camp (3)	Foreign outside camp (4)
Panel A: References				
Quality of references	Aver. num. of cits. to recent research	Average citation shares to recent research		
All references	2.593	0.686	0.159	0.150
Top 1% references	0.207	0.054	0.012	0.013
Top 3% references	0.479	0.126	0.027	0.029
Top 5% references	0.702	0.181	0.041	0.040
Panel B: Standarized LSA title similarity				
		Average LSA title similarity to recent papers		
Similarity to most similar title		0.376	−0.300	−0.076
Avg. similarity to 5 most similar titles		0.462	−0.399	−0.063

Notes. In Panel A, column (1), the table reports the number of references to recent papers (published between year t-4 and t, where t is the publication year of the citing paper). In Panel A, columns (2)-(4), the table reports citation shares to research from home, foreign countries inside the camp, and foreign countries outside the camp. For top x% references, citation shares are computed over all references (both high quality and low quality). In Panel B, columns (2)-(4), the table reports the standardized (i.e., mean 0 and standard deviation 1) LSA title similarity to papers by scientists from home, foreign countries inside the camp, and foreign countries outside the camp. In the penultimate row, LSA title similarity is computed as the similarity to the most similar title from each camp. In the last row, LSA title similarity is computed as the average similarity to the five most similar titles from each camp. We compute title similarity to recent papers (published between year t-4 and t). The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

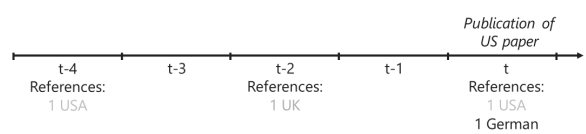

FIGURE III

Example Citing Paper and References

5.6 Figure III: Example citing paper and references

Read:

- This is the mechanical example that turns raw reference lists into the citation-share variables used in the regressions.
- It makes the regressions easy to interpret: the paper is literally comparing how the same citing paper allocates references across home, inside-camp foreign, and outside-camp foreign research.

5.7 Table IV: Changes in international citations

Read:

- The main coefficient is large: citations to recent research from outside the camp fall by about 0.217 relative to home after 1914.
- Citations to recent research from inside the camp also fall, but much less: about 0.072 in the basic specification.
- Splitting the post-1914 period shows the strongest relative collapse during the boycott years, with outside-camp citations down about 0.245.
- This table is the core first-stage evidence that international knowledge flows were disrupted.

TABLE IV

CHANGES IN INTERNATIONAL CITATIONS

Dependent variable: *Citation Shares to recent research*	(1)	(2)	(3)	(4)
Foreign outside camp × post-1914	−0.217***	−0.261***		
	(0.033)	(0.040)		
Foreign outside camp × WWI			−0.222***	−0.229***
			(0.025)	(0.034)
Foreign outside camp × boycott			−0.245***	−0.258***
			(0.034)	(0.052)
Foreign outside camp × post boycott			−0.194***	−0.213***
			(0.042)	(0.051)
Foreign inside camp × post-1914	−0.072*	−0.155***		
	(0.041)	(0.051)		
Foreign inside camp × WWI			−0.111***	−0.148***
			(0.040)	(0.045)
Foreign inside camp × boycott			−0.089**	−0.164***
			(0.042)	(0.057)
Foreign inside camp × post boycott			−0.048	−0.154**
			(0.048)	(0.059)
Paper fixed effects	Yes	Yes	Yes	Yes
Camp fixed effects	Yes	Yes	Yes	Yes
Foreign in/outside time trends		Yes		Yes
Observations	105,378	105,378	105,378	105,378
Number of citing papers	35,126	35,126	35,126	35,126
Within R-squared	0.334	0.335	0.335	0.335

Notes. Each column reports one set of parameter estimates of regression (1) for citing papers published between 1905 and 1930. The dependent variable measures citation shares to research by scientists from home, foreign countries inside the camp, and foreign countries outside the camp. We count citations to recent research, that is, research published in the preceding five years, for example, 1901–1905 for citing papers published in 1905, 1902–1906 for citing papers published in 1906, and so on. The reference/omitted category is the citation share to research from home. Standard errors are clustered at the country-times-field level. Significance levels: *** $p < .01$, ** $p < .05$, and * $p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

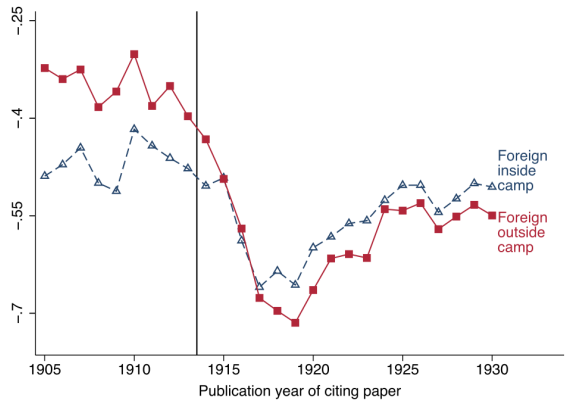

FIGURE IV

International Citation Shares Relative to Home

The figure plots parameter estimates of regression (2). The "Foreign outside camp" line reports point estimates (ω_τ) that measure citation shares to research from outside the camp, relative to research from home. The "Foreign inside camp" line reports point estimates (ι_τ) that measure citation shares to research from foreign scientists inside the camp, relative to research from home. We count citations to recent research, that is, research published in the preceding five years. For example, the first dot (1905) measures relative citation shares to research published between 1901 and 1905. The second dot (1906) measures relative citation shares to research published between 1902 and 1906, and so on. Point estimates and corresponding standard errors are reported in Online Appendix Table A.5. All point estimates are significantly different from 0 at the 1% level. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

5.8 Figure IV: International citation shares relative to home

Read:

- Before the war, there is already home bias. The outside-camp series is around -0.35 relative to home, and the inside-camp series around -0.47.
- After 1914 both series fall sharply, especially the outside-camp series, which drops to roughly -0.71 around the end of the war and early boycott.
- There is partial recovery in the 1920s, but the prewar level is not restored quickly.
- The figure makes the timing of the collapse visually transparent.

5.9 Table V and Figure V: Citations to frontier research

Read:

- Citations to the best foreign research collapse too, not just citations to average work.
- For top 5% research, outside-camp citations fall by about 0.053 relative to home; for top 3%, by about 0.035; for top 1%, by about 0.021.
- Because these frontier citation shares are very small to begin with, the implied percentage declines are huge: about 95% for top 5% and top 3%, and 131% for top 1%.
- The inside-camp declines are smaller than the outside-camp declines.
- This is one of the paper's most important substantive results: access to the narrow frontier is exactly what is being lost.

TABLE V

CHANGES IN INTERNATIONAL CITATIONS: FRONTIER RESEARCH

Dependent variable:	Frontier: 5%		Frontier: 3%		Frontier: 1%	
Cit. Sh. to recent frontier research	(1)	(2)	(3)	(4)	(5)	(6)
Foreign outside camp × post-1914	−0.053***	−0.097***	−0.035***	−0.066***	−0.021***	−0.039***
	(0.017)	(0.021)	(0.013)	(0.013)	(0.006)	(0.007)
Foreign inside camp × post-1914	−0.023	−0.071***	−0.019*	−0.049***	−0.013**	−0.033***
	(0.015)	(0.021)	(0.011)	(0.013)	(0.006)	(0.007)
Paper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Camp fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Nonfrontier research interactions	Yes	Yes	Yes	Yes	Yes	Yes
Foreign *in/outside* time trends		Yes		Yes		Yes
Observations	210,756	210,756	210,756	210,756	210,756	210,756
Number of citing papers	35,126	35,126	35,126	35,126	35,126	35,126
Within R-squared	0.235	0.235	0.299	0.300	0.400	0.400

Notes. Each column reports one set of parameter estimates of regression (1) for citing papers published between 1905 and 1930. The dependent variable measures citation shares to frontier and nonfrontier research by scientists from home, foreign countries inside the camp, and foreign countries outside the camp, that is, six shares for each citing paper. The table only reports estimates for frontier research, although the regressions control for nonfrontier times post-1914 indicators. For the results reported in columns (1)–(2), frontier research is defined as research that ended up in the top 5% of the subject-level citation distribution, counting citations until today. Similarly, for the results reported in columns (3)–(4) (and (5)–(6)), frontier research is defined as research that ended up in the top 3% (and 1%) of the subject-level citation distribution. We count citations to recent research, that is, research published in the preceding five years: e.g., 1901–1905 for citing papers published in 1905, 1902–1906 for citing papers published in 1906, and so on. The reference/omitted category is the citation share to frontier research from home. Standard errors are clustered at the country-times-field level. Significance levels: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

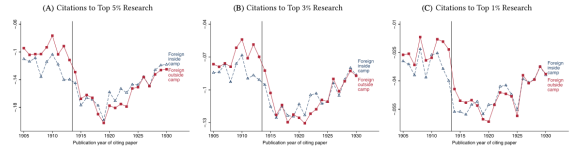

FIGURE V

International Citation Shares Relative to Home: Citations to High-Quality Research

Each panel plots one set of parameter estimates of a variant of regression (2). For the results reported in Panel A, we split citation shares to research from home into research that ended up in the top 5% of the citation distribution and research that ended up in the bottom 95%. Similarly, we split citation shares to research produced inside the camp and outside the camp. The "Foreign outside camp" line measures citation shares to top 5% research from outside the camp, relative to top 5% research from home. The "Foreign inside camp" line measures citation shares to top 5% research from foreign scientists inside the camp, relative to top 5% research from home. The regressions also include the citation shares to nonfrontier research from outside the camp, inside the camp, and home. In all panels, we count citations to recent research, that is, research published in the preceding five years. For example, the first dot (1905) measures relative citation shares to top 5% research published between 1901 and 1905, and so on. Panel B reports estimates for changes in citation shares to research that ended up in the top 3% of the citation distribution, and Panel C reports estimates for changes in citation shares to research that ended up in the top 1% of the citation distribution. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

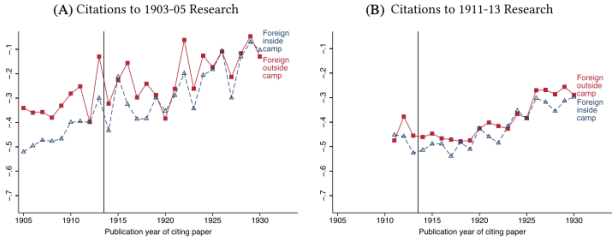

FIGURE VI

International Citation Shares Relative to Home: Citations to Prewar Research

Each panel plots one set of parameter estimates of regression (2) with citation shares to prewar research as the dependent variable. Differently from previous figures, each dot of any line measures relative citation shares to a fixed cohort of research, published either in 1903–1905 (Panel A) or in 1911–1913 (Panel B). In Panel A, the "Foreign outside camp" line reports point estimates (ω_τ) that measure citation shares to 1903–1905 research from outside the camp, relative to 1903–1905 research from home. The "Foreign inside camp" line reports point estimates (ι_τ) that measure citation shares to 1903–1905 research from inside the camp, relative to 1903–1905 research from home. Panel B refers to research published in 1911–1913 and can be interpreted analogously. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

5.10 Figure VI: Citations to prewar research

Read:

- Relative citations to older foreign research do *not* collapse after 1914.
- That pattern supports the interpretation that scientists still knew the older work but lost access to *new* foreign research.
- This figure is the paper's main argument against a pure political-citation story.

5.11 Table VI and Figure VII: Title similarity

Read:

- Using the most-similar-title measure, title similarity to papers from outside the camp falls by about 0.467 standard deviations relative to home after 1914.

TABLE VI

THE SIMILARITY OF PAPERS AS MEASURED BY LATENT SEMANTIC ANALYSIS

Dependent variable:	Most similar title				Average five most similar titles			
LSA title similarity to recent papers	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Foreign outside camp × post-1914	−0.467***	−0.557***			−0.608***	−0.678***		
	(0.095)	(0.126)			(0.131)	(0.165)		
Foreign outside camp × WWI			−0.459***	−0.517***			−0.575***	−0.639***
			(0.084)	(0.117)			(0.118)	(0.143)
Foreign outside camp × boycott			−0.530***	−0.648***			−0.678***	−0.806***
			(0.111)	(0.193)			(0.150)	(0.213)
Foreign outside camp × post boycott			−0.426***	−0.593***			−0.569***	−0.753***
			(0.099)	(0.188)			(0.134)	(0.192)
Foreign inside camp × post-1914	0.060	−0.164			0.058	−0.226		
	(0.151)	(0.140)			(0.191)	(0.178)		
Foreign inside camp × WWI			−0.019	−0.181			−0.054	−0.267*
			(0.137)	(0.117)			(0.177)	(0.141)
Foreign inside camp × boycott			0.006	−0.320**			0.005	−0.425***
			(0.161)	(0.152)			(0.202)	(0.160)
Foreign inside camp × post boycott			0.122	−0.343**			0.131	−0.482***
			(0.154)	(0.165)			(0.194)	(0.165)
Paper fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Camp fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Foreign in/outside time trends		Yes		Yes		Yes		Yes
Observations	71,586	71,586	71,586	71,586	71,586	71,586	71,586	71,586
Number of citing papers	23,862	23,862	23,862	23,862	23,862	23,862	23,862	23,862
Within R-squared	0.156	0.157	0.157	0.158	0.225	0.228	0.227	0.228

Notes. Each column reports one set of parameter estimates of regression (1) for papers published between 1905 and 1930. The dependent variable measures the standardized (i.e., mean 0 and standard deviation 1) LSA title similarity to papers by scientists from home, foreign countries inside the camp, and foreign countries outside the camp. In columns (1) to (4), LSA title similarity is computed as the similarity to the most similar paper from each camp. In columns (5) to (8), LSA title similarity is computed as the average similarity to the five most similar papers from each camp. We compute title similarity to recent papers, that is, papers published in the preceding five years: 1901–1905 for papers published in 1905, and so on. The reference/omitted category is the LSA title similarity to papers from home. Standard errors are clustered at the country-times-field level. Significance levels: ***$p < .1$, **$p < .05$, and *$p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

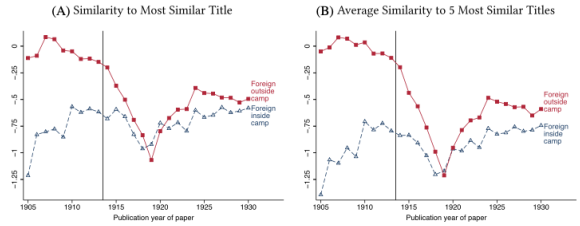

FIGURE VII

International Title Similarity Relative to Home

Each panel plots one set of parameter estimates of the equivalent of regression (2) where the dependent variable measures the standardized (i.e., mean 0 and standard deviation 1) LSA title similarity to papers by scientists from home, foreign countries inside the camp, and foreign countries outside the camp. In Panel A, LSA title similarity is computed as the similarity to the most similar title from each camp. In Panel B, LSA title similarity is computed as the average similarity to the five most similar titles from each camp. The "Foreign outside camp" line reports point estimates (ω_τ) that measure the LSA title similarity to papers from outside the camp, relative to papers from home. The "Foreign inside camp" line reports point estimates (ι_τ) that measure the LSA title similarity to papers from foreign scientists inside the camp, relative to papers from home. We measure title similarity to recent papers, that is, papers published in the preceding five years. For example, the first dot (1905) measures relative title similarity to papers published between 1901 and 1905, and so on. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

- With the five-most-similar-title measure, the fall is even larger, around 0.608 standard deviations.
- Similarity to inside-camp papers changes much less.
- These results confirm the citation evidence with a completely different text-based measure: research programs across enemy camps diverge.

5.12 Figure VIII: Prewar citations to research from home, inside camp, and outside camp

Read:

- This figure shows the cross-field and cross-country exposure variation that identifies the second-stage productivity design.
- The paper's favorite intuitive comparison is U.S. biochemistry versus U.S. biology. U.S. biochemistry relies much more on frontier research from outside the camp, especially Germany; U.S. biology relies more on home frontier research.
- The figure makes it clear that exposure is highly heterogeneous across field-country pairs, which is exactly what the difference-in-differences design needs.

5.13 Table VII: Summary statistics for scientist productivity

Read:

- The scientist panel is large and sparse: the average scientist publishes only 0.267 papers per year.
- Path-breaking and technology-related outcomes are much rarer: Nobel-nominated papers average 0.001 per year, and novel-word introduction averages 0.041 per year.
- The low unconditional frequencies explain why large percentage effects can correspond to small absolute probability changes.

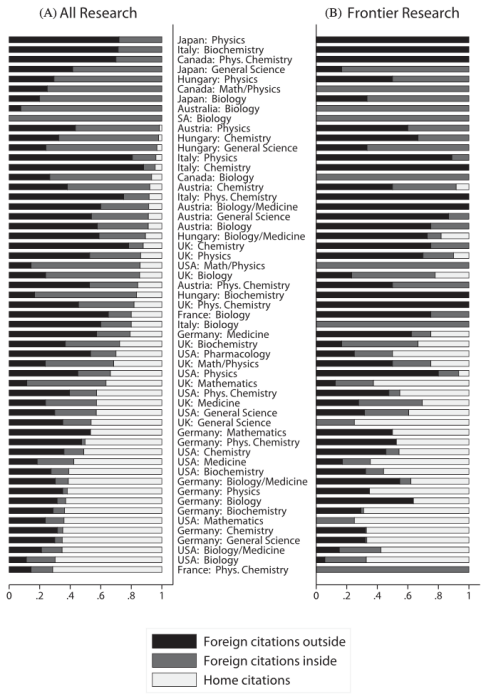

FIGURE VIII

Prewar Citations to Research from Home, Foreign Inside Camp, and Outside Camp

Panel A shows the prewar reliance on all research (i.e., both frontier and non-frontier) from home, from abroad outside the camp, and from abroad inside the camp for each field-country pair. Prewar reliance on all research is calculated as the average citation shares to recent research from home, foreign countries inside the camp, and foreign countries outside the camp for all citing papers published by all university scientists in each field-country pair between 1900 to 1913. Panel B focuses on prewar reliance on frontier research, measured as average shares of citations to top 3% research. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

TABLE VII

SUMMARY STATISTICS: PRODUCTIVITY OF SCIENTISTS

	Mean	Std. dev.
Number of scientists	8,734	
Number of scientist-year observations	227,084	
Career age in years	7.444	7.708
Publications per year	0.267	0.950
Nobel-nominated papers per year	0.001	0.029
Nomination-weighted Nobel-nominated papers per year	0.003	0.152
Number of novel words (word innovation) per year	0.041	0.268
Patent-relevant word innovation per year	0.427	3.538

Notes. The table reports summary statistics for the panel of scientists with a university position by 1914. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt*, publication and citation data from *ISI Web of Science*, Nobel nomination and award data from Nobelprize.org (2014), and patent data from U.S. Patent Office (see Section III for details).

5.14 Table VIII: Effect on publications

Read:

- The preferred publication result is in the top 1% frontier specification: reliance on frontier research from outside the camp carries a coefficient of -1.727 after 1914.
- The corresponding coefficient for frontier research from inside the camp is also negative but much smaller and statistically insignificant.
- The effect becomes smaller as the frontier is broadened from top 1% to top 3% to top 5%.
- This is the main quantitative evidence that the top of the frontier matters disproportionately.

TABLE VIII
EFFECT ON PUBLICATIONS

Dependent variable: *Number of publications*	Frontier: 1% (1)	Frontier: 3% (2)	Frontier: 5% (3)
Prewar reliance on frontier OUT × post-1914	−1.727***	−0.784***	−0.380*
	(0.638)	(0.282)	(0.220)
Prewar reliance on frontier IN × post-1914	−0.827	−0.363	−0.152
	(0.736)	(0.283)	(0.218)
Scientist fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Prewar reliance on nonfrontier	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes
Observations	227,084	227,084	227,084
Number of scientists	8,734	8,734	8,734
Within R-squared	0.062	0.062	0.062

Notes. Each column reports one set of parameter estimates of regression (3) for the panel of university scientists. The dependent variable measures the yearly number of publications in the 160 top journals in our data for the years 1905 to 1930. The number of publications is normalized by the number of authors and standardized to mean zero and standard deviation one within fields. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: ***$p < .01$, **$p < .05$, and *$p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

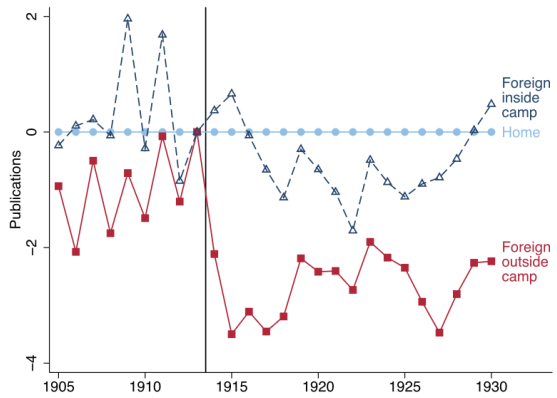

FIGURE IX
Effect on Publications

The figure plots parameter estimates from regression (4). The "Foreign outside camp" line reports point estimates ($\beta_{1\tau}$) that measure changes in yearly publications for scientists in field-country pairs that, in the prewar period, relied on frontier research from outside the camp, compared to scientists who relied on frontier research from home. The "Foreign inside camp" line reports point estimates ($\beta_{2\tau}$) that measure changes in yearly publications for scientists in field-country pairs that, in the prewar period, relied on frontier research from foreign scientists inside the camp, compared to scientists who relied on frontier research from home. Prewar reliance on frontier research is measured by prewar citations to frontier research at the field-country pair level. Frontier research is defined as research that ended up in the top 1% of the subject-level citation distribution, counting citations until today. The regression also controls for prewar reliance on nonfrontier research from each camp interacted with year indicators. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

5.15 Figure IX: Event-study effect on publications

Read:

- There is no negative pretrend before 1914 for outside-camp frontier exposure; if anything, the exposed fields improve slightly before the war.
- After 1914, the outside-camp-exposed line drops sharply and stays low.
- The inside-camp-exposed line also dips, but less and less persistently.
- This figure is crucial because it makes the dynamic treatment pattern and lack of bad pretrends easy to see.

5.16 Table IX: Publication robustness checks

Read:

- The publication result survives removal of author normalization and the addition of camp-year, field-year, and camp-field-year fixed effects.
- In the most saturated camp-field-year specification, the top 1% outside-camp coefficient remains around -1.667, and the top 3% outside-camp coefficient becomes even more negative at -1.105.
- The design is therefore not driven by broad common shocks to one camp or one discipline.

TABLE IX

EFFECT ON PUBLICATIONS: ROBUSTNESS CHECKS

Dependent variable: *Number of publications*	# pub. per author (1)	# pub. not normal. by # authors (2)	Control for camp × year (3)	Control for field × year (4)	Control for camp × field × year (5)
Panel A: Frontier measured by top 1%					
Prewar reliance on 1% frontier OUT × post-1914	−1.727***	−1.775**	−1.489*	−1.655***	−1.667***
	(0.638)	(0.669)	(0.826)	(0.616)	(0.491)
Prewar reliance on 1% frontier IN × post-1914	−0.827	−0.923	−0.823	−0.782	−0.762
	(0.736)	(0.730)	(0.764)	(0.725)	(0.698)
Within R-squared	0.062	0.066	0.064	0.064	0.068
Panel B: Frontier measured by top 3%					
Prewar reliance on 3% frontier OUT × post-1914	−0.784***	−0.813***	−0.596	−0.744**	−1.105***
	(0.282)	(0.288)	(0.379)	(0.295)	(0.237)
Prewar reliance on 3% frontier IN × post-1914	−0.363	−0.454	−0.432	−0.300	−0.311
	(0.283)	(0.279)	(0.297)	(0.292)	(0.265)
Within R-squared	0.062	0.066	0.063	0.064	0.068
Panel C: Frontier measured by top 5%					
Prewar reliance on 5% frontier OUT × post-1914	−0.380*	−0.400*	−0.224	−0.372	−0.686***
	(0.220)	(0.222)	(0.262)	(0.270)	(0.256)
Prewar reliance on 5% frontier IN × post-1914	−0.152	−0.205	−0.170	−0.120	−0.167
	(0.218)	(0.218)	(0.225)	(0.230)	(0.192)
Within R-squared	0.062	0.066	0.063	0.064	0.068
Scientist fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes			
Prewar reliance on nonfrontier	Yes	Yes	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes	Yes	Yes
Camp × year fixed effects			Yes		
Field × year fixed effects				Yes	
Camp × field × year fixed effects					Yes
Observations	227,084	227,084	227,084	227,084	227,084
Number of scientists	8,734	8,734	8,734	8,734	8,734

Notes. Each column and each panel reports one set of parameter estimates of regression (3) for the panel of university scientists. For the results presented in Panel A (B and C), we measure the frontier as the top 1% (top 3% and top 5%) research. The dependent variable measures the yearly number of publications in the 160 top journals in our data for the years 1905 to 1930. In column (1) and (3)–(5), the dependent variable is normalized by the number of authors and standardized to mean 0 and standard deviation 1 within fields. In column (2), the dependent variable is not normalized by the number of authors but standardized to mean 0 and standard deviation 1 within fields. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

TABLE X

EFFECT ON PUBLICATIONS: DISRUPTION OF KNOWLEDGE FLOWS OR OTHER WAR RELATED DISRUPTION?

Dependent variable: *Number of publications*	Control for combat in country (1)	Control for total deaths per capita (2)	Control for civilian deaths per capita (3)	Control for (1), (2), and (3) (4)	Exclude chemistry (5)	# pub. in own-camp journals (6)
Panel A: Frontier measured by top 1%						
Prewar reliance on 1% frontier OUT × post-1914	−1.846***	−1.547***	−1.987***	−1.649***	−1.721***	−1.484**
	(0.527)	(0.414)	(0.433)	(0.385)	(0.563)	(0.639)
Prewar reliance on 1% frontier IN × post-1914	−0.739	−0.351	−0.489	−0.248	−0.653	−0.545
	(0.731)	(0.742)	(0.690)	(0.666)	(0.740)	(0.707)
Within R-squared	0.068	0.068	0.068	0.069	0.069	0.068
Panel B: Frontier measured by top 3%						
Prewar reliance on 3% frontier OUT × post-1914	−1.158***	−0.984***	−1.297***	−1.056***	−1.142***	−1.074***
	(0.232)	(0.241)	(0.179)	(0.195)	(0.271)	(0.314)
Prewar reliance on 3% frontier IN × post-1914	−0.297	0.024	−0.109	0.089	−0.281	−0.214
	(0.272)	(0.312)	(0.268)	(0.278)	(0.281)	(0.301)
Within R-squared	0.068	0.068	0.068	0.069	0.069	0.068
Panel C: Frontier measured by top 5%						
Prewar reliance on 5% frontier OUT × post-1914	−0.663**	−0.625**	−0.888***	−0.676**	−0.632**	−0.566*
	(0.280)	(0.270)	(0.221)	(0.266)	(0.296)	(0.339)
Prewar reliance on 5% frontier IN × post-1914	−0.176	−0.098	−0.061	−0.086	−0.060	−0.078
	(0.191)	(0.205)	(0.177)	(0.189)	(0.202)	(0.212)
Within R-squared	0.068	0.068	0.068	0.068	0.069	0.068
Scientist fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Prewar reliance on nonfrontier	Yes	Yes	Yes	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes	Yes	Yes	Yes
Camp × field × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	227,084	227,084	227,084	227,084	197,782	215,046
Number of scientists	8,734	8,734	8,734	8,734	7,607	8,271

Notes. Each column and section reports one set of parameter estimates of regression (3) for the panel of university scientists. In columns (1)–(5), the dependent variable measures the yearly number of publications in all 160 top journals in our data in the years 1905 to 1930. In column (6), the dependent variable measures the yearly number of publications in own-camp journals in our data in the years 1905 to 1930. In all columns, the dependent variable is normalized by the number of authors and standardized to mean 0 and standard deviation 1 within fields. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .1$. The data were collected by the authors and combine: scientist census data from *Minerva—Handbuch der Gelehrten Welt*, publication and citation data from *ISI Web of Science*, and war intensity data from Mouguel (2011), *1914–1918 online: International Encyclopedia of the First World War*, and Wikipedia (see Section III and Online Appendix E.5 for details).

5.17 Table X: War-disruption robustness checks

Read:

- Controlling for combat on own territory, total deaths per capita, or civilian deaths per capita does not erase the result.
- Excluding chemists does not erase the result.
- Restricting to own-camp journals does not erase the result.
- These are exactly the robustness checks one wants against the critique that this is just "general war disruption" rather than knowledge-flow disruption.

5.18 Figure X: Within-U.S. variation

Read:

- Restricting to U.S. scientists only still shows a sharp post-1914 relative decline in fields that relied on frontier research from outside the camp.
- This matters because the war was not fought on U.S. territory, and it reduces concern that the main results are driven by country-level wartime destruction.
- It also shows that the design is not just picking up the broad rise of U.S. science.

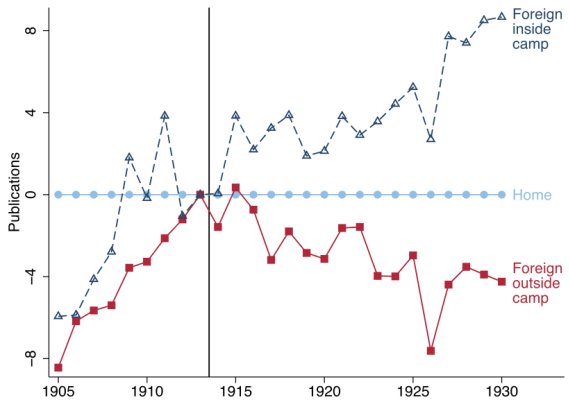

FIGURE X

Effect on Publications: Within U.S. Variation

The figure plots parameter estimates from regression (4) when we restrict the sample to scientists based in the United States. The "Foreign outside camp" line reports point estimates ($\beta_{1\tau}$) that measure changes in yearly publications for scientists in field-country pairs that, in the prewar period, relied on frontier research from outside the camp, compared to scientists who relied on frontier research from home. The "Foreign inside camp" line reports point estimates ($\beta_{2\tau}$) that measure changes in yearly publications for scientists in field-country pairs that, in the prewar period, relied on frontier research from foreign scientists inside the camp, compared to scientists who relied on frontier research from home. Prewar reliance on frontier research is measured by prewar citations to frontier research at the field-country pair level. Frontier research is defined as research that ended up in the top 1% of the subject-level citation distribution, counting citations until today. The regression also controls for prewar dependence on nonfrontier research from each camp interacted with year indicators. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt* and publication and citation data from *ISI Web of Science* (see Section III for details).

TABLE XI

EFFECT ON NOBEL-NOMINATED PAPERS

Dependent variable:	Nomination paper		Nomination paper weighted by # noms.	
	(1)	(2)	(3)	(4)
Panel A: Frontier measured by top 1%				
Prewar reliance on 1% frontier OUT	−0.021**	−0.019*	−0.148***	−0.175*
× post-1914	(0.008)	(0.011)	(0.052)	(0.103)
Prewar reliance on 1% frontier IN	−0.005	−0.003	−0.048	−0.033
× post-1914	(0.008)	(0.009)	(0.041)	(0.062)
Within R-squared	0.001	0.005	0.001	0.004
Panel B: Frontier measured by top 3%				
Prewar reliance on 3% frontier OUT	−0.012***	−0.012***	−0.061**	−0.073
× post-1914	(0.004)	(0.004)	(0.028)	(0.045)
Prewar reliance on 3% frontier IN	−0.005	−0.006	−0.021	−0.011
× post-1914	(0.004)	(0.005)	(0.018)	(0.020)
Within R-squared	0.001	0.005	0.001	0.004
Panel C: Frontier measured by top 5%				
Prewar reliance on 5% frontier OUT	−0.010**	−0.010**	−0.072**	−0.074**
× post-1914	(0.004)	(0.004)	(0.027)	(0.030)
Prewar reliance on 5% frontier IN	−0.002	−0.003	0.012	0.020
× post-1914	(0.003)	(0.003)	(0.017)	(0.019)
Within R-squared	0.001	0.005	0.001	0.004
Scientist fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes		Yes	
Prewar reliance on nonfrontier	Yes	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes	Yes
Camp × field × year fixed effects		Yes		Yes
Observations	227,084	227,084	227,084	227,084
Number of scientists	8,734	8,734	8,734	8,734

Notes. Each column and each panel reports one set of parameter estimates of regression (3) for the panel of university scientists. The dependent variable in columns (1) and (2) is an indicator that equals 1 if a scientist published a Nobel-nominated paper in a certain year between 1905 and 1930, and 0 for all other years. The dependent variable in columns (3) and (4) weights the Nobel-nominated paper indicator by the number of nominations in the two years before a candidate's last nomination. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: *** $p < .01$, ** $p < .05$, and * $p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt*, publication and citation data from *ISI Web of Science*, and Nobel nomination data from Nobelprize.org (2014) (see Section III for details).

5.19 Table XI: Effect on Nobel-nominated papers

Read:

- In the top 1% frontier specification, outside-camp reliance reduces the probability of producing a Nobel-nominated paper after 1914 by about 0.021 in the normalized regression metric.
- In the paper's economic interpretation, for a field like U.S. biochemistry the implied probability decline is about 0.001, roughly equal to the entire prewar probability of producing a Nobel-nominated paper.
- The effect is weaker or absent for inside-camp frontier reliance.
- This table moves the result from quantity of papers to path-breaking scientific quality.

5.20 Table XII: Effect on word innovation and patent-relevant words

Read:

- In the top 1% frontier specification, outside-camp reliance strongly reduces novel-word creation and also reduces patent-relevant words.
- The inside-camp effect is also negative, especially in the top 1% frontier case.
- The patent-relevant measure is especially interesting because it links basic scientific vocabulary to later applied technology.
- Example: "magnetron" later appears 9,538 times in 997 patents; "electroencephalogram" later appears seven times in three patents.

TABLE XII

Effect on Word Innovation and Patents

Dependent variable:	Novel scientific words		Patent-relevant words	
	(1)	(2)	(3)	(4)
Panel A: Frontier measured by top 1%				
Prewar reliance on 1% frontier OUT	−1.229***	−0.778**	−1.134***	−1.207***
× post-1914	(0.441)	(0.327)	(0.349)	(0.297)
Prewar reliance on 1% frontier IN	−0.910*	−0.918**	−0.569*	−0.661**
× post-1914	(0.468)	(0.368)	(0.295)	(0.253)
Within R-squared	0.025	0.028	0.015	0.018
Panel B: Frontier measured by top 3%				
Prewar reliance on 3% frontier OUT	−0.311	−0.359**	−0.415**	−0.639***
× post-1914	(0.261)	(0.173)	(0.198)	(0.155)
Prewar reliance on 3% frontier IN	−0.149	−0.181	−0.115	−0.129
× post-1914	(0.225)	(0.167)	(0.182)	(0.139)
Within R-squared	0.024	0.028	0.015	0.018
Panel C: Frontier measured by top 5%				
Prewar reliance on 5% frontier OUT	−0.182	−0.298*	−0.339**	−0.542***
× post-1914	(0.204)	(0.164)	(0.158)	(0.149)
Prewar reliance on 5% frontier IN	−0.149	−0.150	−0.136	−0.161
× post-1914	(0.173)	(0.134)	(0.134)	(0.102)
Within R-squared	0.024	0.028	0.015	0.018
Scientist fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes		Yes	
Prewar reliance on nonfrontier	Yes	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes	Yes
Camp × field × year fixed effects		Yes		Yes
Observations	227,084	227,084	227,084	227,084
Number of scientists	8,734	8,734	8,734	8,734

Notes. Each column and each panel reports one set of parameter estimates of regression (3) for the panel of university scientists. The dependent variable in columns (1) and (2) counts the number of novel words that appeared in the title of a scientific paper published in year t. The dependent variable in columns (3) and (4) counts the number of times each of the novel words (as defined above) was used in the text of any patent granted by the U.S. Patent Office in years $t + 15$ and $t + 30$. The dependent variable in columns (3) and (4) is winsorized at the 99th percentile. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt*, publication and citation data from *ISI Web of Science*, Nobel nomination and award data from Nobelprize.org (2014), and patent data from the *U.S. Patent Office* (see Section III for details).

TABLE XIII

Effect on Productivity by Prewar Quality of Scientists

	⩽ median (1)	> median (2)	⩽ median (3)	> median (4)	⩽ median (5)	> median (6)	⩽ median (7)	> median (8)	⩽ median (9)	> median (10)
Dependent variable:	Number of publications		Nomination paper		Nom. paper (weighted)		Novel scientific words		Patent-relevant words	
Panel A: Frontier measured by top 1%										
Prewar reliance on 1% frontier OUT	−0.430*	−3.296***	0.004	−0.052**	−0.023	−0.373*	−0.149	−1.656**	−0.465***	−2.163***
× post-1914	(0.227)	(1.237)	(0.007)	(0.022)	(0.039)	(0.219)	(0.099)	(0.778)	(0.135)	(0.714)
Prewar reliance on 1% frontier IN	0.028	−1.693	−0.000	−0.010	−0.004	−0.083	0.026	−1.884***	−0.134	−1.148*
× post-1914	(0.500)	(1.046)	(0.006)	(0.017)	(0.030)	(0.125)	(0.167)	(0.528)	(0.187)	(0.587)
Within R-squared	0.124	0.077	0.008	0.009	0.016	0.006	0.025	0.046	0.014	0.031
Panel B: Frontier measured by top 3%										
Prewar reliance on 3% frontier OUT	−0.278**	−2.131***	0.001	−0.031***	−0.004	−0.166	−0.164***	−0.540	−0.327***	−0.992***
× post-1914	(0.120)	(0.569)	(0.002)	(0.011)	(0.013)	(0.100)	(0.056)	(0.347)	(0.072)	(0.300)
Prewar reliance on 3% frontier IN	−0.088	−0.668	−0.001	−0.012	−0.004	−0.016	−0.037	−0.421	−0.136	−0.125
× post-1914	(0.182)	(0.438)	(0.003)	(0.008)	(0.013)	(0.034)	(0.073)	(0.294)	(0.096)	(0.290)
Within R-squared	0.124	0.077	0.008	0.009	0.016	0.006	0.025	0.046	0.014	0.031
Panel C: Frontier measured by top 5%										
Prewar reliance on 5% frontier OUT	−0.070	−1.286***	−0.001	−0.024**	−0.023	−0.142**	−0.140**	−0.295	−0.290***	−0.714***
× post-1914	(0.167)	(0.454)	(0.003)	(0.010)	(0.015)	(0.065)	(0.067)	(0.306)	(0.079)	(0.266)
Prewar reliance on 5% frontier IN	−0.057	−0.392	−0.000	−0.008	0.001	0.036	−0.057	−0.335	−0.151**	−0.214
× post-1914	(0.150)	(0.293)	(0.002)	(0.005)	(0.010)	(0.036)	(0.055)	(0.214)	(0.064)	(0.178)
Within R-squared	0.124	0.077	0.008	0.009	0.016	0.006	0.025	0.046	0.014	0.031
Scientist fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Prewar reliance on nonfrontier	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Career age × field interactions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Camp × field × year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	121,446	105,638	121,446	105,638	121,446	105,638	121,446	105,638	121,446	105,638
Number of scientists	4,671	4,063	4,671	4,063	4,671	4,063	4,671	4,063	4,671	4,063

Notes. Each column and each panel reports one set of parameter estimates of regression (3) for the panel of university scientists. We divide the sample of scientists into two subsamples: scientists who published less or the same number of papers than the median scientist (results reported in odd columns) and scientists who published more papers than the median scientist (even columns). The median number of papers is calculated at the field level for the period 1905–1913. The dependent variables are defined as in the previous tables. "Prewar reliance on frontier OUT" is the prewar citation share to frontier research (1%, 3%, or 5%) from outside the camp. "Prewar reliance on frontier IN" is the prewar citation share to frontier research (1%, 3%, or 5%) from foreign countries inside the camp. The reference/omitted category is "Prewar reliance on frontier HOME." Standard errors are clustered at the country-times-field level. Significance levels: $^{***}p < .01$, $^{**}p < .05$, and $^{*}p < .1$. The data were collected by the authors and combine scientist census data from *Minerva—Handbuch der Gelehrten Welt*, publication and citation data from *ISI Web of Science*, Nobel nomination data from Nobelprize.org, and patent data from the U.S. Patent Office (see Section III for details).

5.21 Table XIII: Effect by prewar quality of scientists

Read:

- Above-median scientists are hit much harder in absolute terms than below-median scientists.
- For top 1% outside-camp exposure, the publication effect is about -3.295 for above-median scientists versus about -0.430 for below-median scientists.
- Similar patterns appear for Nobel-type outcomes, novel words, and patent-relevant words.
- The paper's interpretation is complementarity: high-quality scientists benefit the most from access to frontier knowledge.

6 Short summary

- **Conclusion.** WWI and the postwar boycott sharply disrupted international scientific exchange, and that disruption reduced both frontier scientific production and the technological usefulness of science.
- **Intuition.** Scientists build on recent frontier work. When the flow of that work from abroad is cut off, the most exposed fields lose one of their key production inputs.
- **Empirical lesson.** The best research matters disproportionately. Access to the top 1% of prior work is more important than access to the top 3% or top 5%.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Historical fact.** WWI and the boycott caused a sharp, measurable decline in international scientific cooperation.
- **Knowledge-flow evidence.** Citations to recent foreign research outside the camp collapsed, while citations to prewar foreign research did not. Paper titles also became less similar across enemy camps.
- **Scientific-output evidence.** Scientists in field-country pairs that relied on frontier research from abroad published fewer papers and fewer Nobel-level papers after 1914.
- **Concept and technology evidence.** Those scientists also introduced fewer new scientific concepts, and fewer of those concepts flowed into later patents.
- **Frontier result.** The top 1% frontier matters much more than broader definitions of the frontier.
- **Heterogeneity result.** The highest-quality scientists are the ones who lose the most, in absolute terms, when access to frontier knowledge is curtailed.

7.2 Economic intuition

- Existing knowledge is an input into idea production, and the very best existing knowledge is a particularly valuable input.
- International scientific exchange matters because no one country or field contains the whole frontier.
- Once scientists are cut off from the latest foreign advances, research programs diverge and productivity falls, especially in fields that had specialized in drawing on the foreign frontier.
- The complementarity with scientist quality is natural: the most able researchers are best positioned to exploit frontier inputs.

7.3 Takeaway

This paper turns a major geopolitical rupture into evidence on the economics of knowledge. Its central message is that scientific progress depends not just on doing more research, but on maintaining

access to the very best research wherever it is produced. When that access breaks down, science becomes more parochial, frontier output falls, and the downstream technological consequences are visible even decades later.